

Comparative Analysis of the Earlier and Current OMINO Models

Executive Summary

The earlier model and the current model describe the same underlying ambition from different stages of development.

The **earlier model** begins with geometry:

- four semantic elements arranged as tetrahedral vertices;
- six pairwise relationships represented as edges;
- CONS treated as a central hinge;
- a six-variable truth-table field with $2^6 = 64$ relationship states;
- higher polytopes used as spatial lenses;
- 256-bit storage words divided into geometric regions;
- Delta used to track changes in relationships.

The **current model** retains that geometry but separates it into more precise layers:

- the tetrahedral incidence object $\langle 4, 6, 4, 1 \rangle$;
- the unconventional OMINO $[6, 4, 2]$ operational gauge;
- FS, GS, RS, and US as the four persistent controls;
- a six-edge truth-table coordinate;
- OMICRON $[7, 4, 3]$ and OMNICRON $[8, 4, 4]$ as integrity extensions;
- exact low/high control planes and tangential rows;
- projection, clock, rotation, and resolution as separate coordinates;
- incidence-derived geometric quantities rather than stored floating-point constants;
- Emergent-Axial-Lisp as the lowering mechanism from symbolic constructions to executable coordinates.

The older model should therefore not be discarded. It should be **translated into the newer vocabulary**, with unsupported claims removed and unfinished mappings marked explicitly.

1. Developmental Lineage

1.1 Earliest logic-first construction

Before the Hamming terminology was introduced, OMINO was conceived as a truth-table logic counter.

Four semantic elements formed the tetrahedral vertices:

$$V = \{\text{RULES, FACTS, CLOSURES, COMBINATORS}\}.$$

Every pair of vertices determines one edge:

$$\binom{4}{2} = 6.$$

The six relations therefore define a six-variable Boolean state:

$$e \in \{0, 1\}^6.$$

The complete relation-state field has:

$$2^6 = 64$$

possible edge configurations.

In this earliest model:

- the four vertices supplied semantic categories;
- the six truth-table bits supplied simultaneous relationships;
- Delta tracked changes in those relationships;
- CONS supplied the coordinate or ordered construction associated with the state.

This is the conceptual origin of the later OMINO [6, 4, 2] notation.

1.2 Geometry-first spatial model

The truth-table construction was then interpreted as a tetrahedral spatial object:

$$\mathcal{T} = (V_4, E_6, F_4, C_1).$$

It contains:

- four vertices;
- six edges;
- four triangular faces;
- one distinguished centroid.

The uploaded geometry development explicitly defines the tetrahedral incidence values 4, 6, 4, 1 and verifies the standard incidence balances.

The intended semantic binding was:

Tetrahedral component	OMI interpretation
Four vertices	RULES, FACTS, CLOSURES, COMBINATORS
Six edges	Direct pairwise transformation channels

Tetrahedral component	OMI interpretation
Four faces	Three-vertex closure contexts
Centroid	CONS ordered-pair hinge

1.3 Hamming-era reinterpretation

The geometry was later expressed through code-like notation:

```
OMINO      [6,4,2]
OMICRON    [7,4,3]
OMNICRON   [8,4,4]
```

The newer specification correctly distinguishes these:

- OMINO `[6,4,2]` is an **architectural incidence profile**, not a conventional named Hamming code.
- OMICRON `[7,4,3]` is the compact Hamming integrity form.
- OMNICRON `[8,4,4]` is the extended integrity form.

Therefore Hamming theory did not originate the six-coordinate OMINO. It supplied a later integrity framework for a preexisting tetrahedral truth-table construction.

2. Terminology Comparison

2.1 OMINO versus OMNION

Earlier form

The older drafts often used:

```
OMNION
OMNION centroid
one unified OMNION footprint
```

Current form

The normalized name is:

```
OMINO
```

OMINO is short for the generalized polyomino/polyomnio construction and refers to the composable Polyharmonic Omino Block.

Resolution

OMNION should be treated as deprecated terminology unless it refers to a separately defined historical object.

Use:

OMINO centroid
OMINO block
Polyharmonic Omino Block

2.2 Receipt terminology

Earlier form

CLOSURES was described as:

Receipt Boundary
Validated System Limits

Current form

The newer architecture rejects receipt or external-authority semantics.

CLOSURES should mean:

finite relation boundary
termination condition
internal closure surface
validated bounded limit

Resolution

The structural role survives, but the word **receipt** should be removed.

2.3 Validation terminology

The older statement that Catalan solids “validate” Archimedean solids has geometric value but requires a narrower definition.

The current meaning should be:

The Catalan dual supplies a reciprocal incidence projection of the corresponding Archimedean construction, allowing the primal and dual forms to be compared from a common centroid.

This is an internal geometric consistency relation, not external certification.

3. Tetrahedral File Architecture

3.1 Earlier claim

The older text described “the tetrahedral geometry of the 15 formal files” and presented five file types:

```
RULES
FACTS
CLOSURES
COMBINATORS
CONS
```

across:

```
.omi
.imo
.o
```

3.2 Current interpretation

The fifteen files are not fifteen vertices or fifteen tetrahedral components.

They are:

$$5 \text{ semantic families} \times 3 \text{ representation surfaces} = 15.$$

Each representation surface contains a tetrahedral semantic view:

```
RULES.ext  
FACTS.ext  
CLOSURES.ext  
COMBINATORS.ext  
CONS.ext
```

where `ext` is `.omi`, `.imo`, or `.o`.

Thus there are three corresponding representations of one semantic tetrahedron:

```
.omi  
    declarative surface  
  
.imo  
    defined or intermediate surface  
  
.o  
    executable finite surface
```

3.3 Preserved insight

The important older insight survives:

The file system is not a flat list. It is a family of corresponding semantic tetrahedra sharing the same four vertices and CONS center.

4. Four Vertices and Six Edges

4.1 Earlier edge masks

The older model supplied six four-bit masks:

```
0x03  
0x05  
0x09  
0x06  
0x0A  
0x0C
```

Every mask has Hamming weight two.

With the normalized bit assignment:

```
bit 0 = RULES
bit 1 = FACTS
bit 2 = CLOSURES
bit 3 = COMBINATORS
```

the masks become:

Mask	Edge
0x03	RULES-FACTS
0x05	RULES-CLOSURES
0x09	RULES-COMBINATORS
0x06	FACTS-CLOSURES
0x0A	FACTS-COMBINATORS
0x0C	CLOSURES-COMBINATORS

This is an exact enumeration of the six edges of K_4 , the complete graph on four vertices.

4.2 Current six-bit relation state

The earlier masks select one edge from the four vertices.

The newer six-bit field represents the simultaneous state of all six edges:

```
edge bit 0 = RULES-FACTS
edge bit 1 = RULES-CLOSURES
edge bit 2 = RULES-COMBINATORS
edge bit 3 = FACTS-CLOSURES
edge bit 4 = FACTS-COMBINATORS
edge bit 5 = CLOSURES-COMBINATORS
```

Therefore two different coordinate types must remain separate:

```
4-bit weight-two mask
  identifies one vertex pair
```

6-bit edge-state word
identifies an arbitrary subset of all six relationships

The latter has 64 possible values.

4.3 What changed

The old model sometimes treated the four-bit pair masks and the six-variable truth table as though they were the same surface.

The current model separates them:

edge index $\rightarrow$ vertex-pair mask

and:

six-bit state $\rightarrow$ set of active edges.

This is a major clarification.

5. OMINO [6, 4, 2]

5.1 Earlier interpretation

The older model implicitly used:

6
tetrahedral relation channels

4
semantic vertex categories

2
paired or reciprocal readings

but this was not consistently distinguished from coding-theory notation.

5.2 Current interpretation

The normalized profile is:

$$\text{OMINO}_{642} = (E_6, C_4, D_2)$$

where:

E_6
six tetrahedral pairwise relationships

C_4
four persistent controls:
FS, GS, RS, US

D_2
two reciprocal classes:
relation/scope
or face/vertex

OMINO $[6, 4, 2]$ is therefore:

A six-coordinate operational gauge derived from four-control tetrahedral incidence and two reciprocal interpretations.

It is not asserted to be a conventional error-correcting code with minimum distance two.

5.3 Relationship to FS, GS, RS, and US

A direct binding can be:

RULES	$\leftrightarrow$ FS
FACTS	$\leftrightarrow$ GS
CLOSURES	$\leftrightarrow$ RS
COMBINATORS	$\leftrightarrow$ US

or another explicitly frozen permutation.

The specific permutation remains a normative choice. What is fixed structurally is that four persistent controls generate six pairwise relations.

6. CONS Centroid

6.1 Earlier model

CONS was described as:

```
absolute center  
centroid hinge  
compression center
```

6.2 Current model

CONS is not a fifth ordinary tetrahedral vertex.

It is a distinguished computational reference:

```
ordered-pair constructor  
CAR/CDR hinge  
branch point  
projective origin  
coordinate binder
```

The correct relation is:

$$\text{CONS}(\text{CAR}, \text{CDR}).$$

CONS must preserve order because:

$$a \oplus b = b \oplus a$$

while generally:

$$\text{CONS}(a, b) / \text{CONS}(b, a).=$$

6.3 What changed

The earlier model sometimes described CONS as compressing an interaction lattice directly into an `.o` file.

The current model requires CONS to preserve:

- CAR/CDR orientation;
- source coordinates;
- selected relation;
- projection profile;
- Delta state;
- integrity information.

It is therefore a constructor and coordinate hinge, not merely a compression operation.

7. Truth Table and Delta

7.1 Earlier purpose

The truth-table counter predated the Hamming layer.

Its purpose was to enumerate all states of the six tetrahedral relationships:

$$2^6 = 64.$$

Delta was intended to track which relationships changed.

The simplest relation-change operation is:

$$\delta_t = e_t \oplus e_{t-1}.$$

7.2 Current Delta kernel

The later formal kernel expands this into a deterministic width-bounded rotate/XOR transition:

$$\Delta_n(x) = \text{mask}_n (\text{rotr}(x, 1) \oplus \text{rotr}(x, 3) \oplus \text{rotr}(x, 2) \oplus C_n).$$

The uploaded development proves deterministic replay for equal seeds and profiles.

7.3 Preserved relationship

The newer Delta does not replace the original truth-table purpose.

The layers are:

six-edge truth-table state

logical relationship configuration

basic XOR difference

identifies changed relationships

bounded rotate/XOR Delta

advances that relationship state through an oriented logical trajectory

replay

regenerates the trajectory deterministically

8. Address-Space Comparison

8.1 Older 256-position geometric division

The older model divided the 256-position word into eight 32-position lenses:

```
0x00-0x1F
0x20-0x3F
0x40-0x5F
0x60-0x7F
0x80-0x9F
0xA0-0xBF
0xC0-0xDF
0xE0-0xFF
```

Each lens was divided as:

$$32 = 24 + 8.$$

The first 24 positions were assigned to a Triakis body, and the final eight to a Stellated terminal.

The final eight were divided further:

$$8 = 4 + 4$$

into:

```
first four
  Regular tetrahedron shared word

last four
  Tangent tetrahedron control word
```

8.2 Newer coding partitions

The newer nested gauge model reads the same 32-position lens through several terminal depths:

$$32 = 24 + 8$$

$$32 = 26 + 6$$

$$32 = 27 + 5$$

$$32 = 28 + 4.$$

Therefore the first lens contains:

```
0x18-0x1F
    eight-position OMNICON terminal

0x1A-0x1F
    six-position OMINO terminal

0x1B-0x1F
    five-position Tetragrammatron terminal

0x1C-0x1F
    four-position Metatron/control terminal
```

8.3 Reconciliation

These are two views of the same terminal.

Geometric view

```
0x18-0x1F
    Stellated terminal

0x18-0x1B
    Regular shared word

0x1C-0x1F
    Tangent control word
```

Coding/gauge view

```
0x18-0x1F
    OMNICON 8

0x1A-0x1F
    OMINO 6

0x1B-0x1F
    Tetragrammatron 5

0x1C-0x1F
    Metatron/control 4
```

The two interpretations can coexist if the active view is explicitly identified.

9. Low and High Control Planes

9.1 Earlier model

The older model treated all eight 32-position lenses similarly and assigned geometric names within each lens.

9.2 Current model

The newer model gives special authority to two antipodal control planes:

```
low affine control plane
  0x00-0x1F

high projective control plane
  0x80-0x9F
```

They are paired through:

$$x_{\text{high}} = x_{\text{low}} \oplus 0x80.$$

The rows:

```
0x20-0x7F
0xA0-0xFF
```

are then interpreted as tangential operational expansion rows.

9.3 What changed

The older model treated the 32-position segments primarily as repeating geometric compartments.

The newer model differentiates:

- control planes;
- projective mirrors;
- low and high tangential rows;
- axial origins;
- tangential completions.

This is a stronger executable addressing model.

10. STDIN, STDOUT, and STDERR

10.1 Earlier form

The earlier Lambda/Fano model proposed:

```
Plane 1 = STDIN
Plane 2 = STDOUT
Block design = STDERR
```

It sometimes described the zero polynomial as meaning the system was “always operating over standard error.”

10.2 Current form

The current interpretation should be antipodal:

```
STDIN
  incidence input plane

STDOUT
  antipodal transformed or update plane

STDERR
  centroid-relative difference field
```

A suitable abstract relation is:

$$\text{STDERR}(i, o) = i \oplus A(o),$$

where A is the declared Azimuth transformation.

Then:

```
STDERR = 0
  the antipodal relation closes
```

STDERR ≠ 0
the difference field identifies non-closure

10.3 Meaning of 0x1D

The formal Delta implementation repeats decimal 29, or hexadecimal 0x1D, across the selected width.

This gives two distinct statuses:

implemented fact
0x1D is the repeated Delta byte

architectural interpretation
0x1D may serve as a standard-difference or STDERR-related carrier

The second interpretation must be explicitly frozen before it is normative.

11. Quadratic Form Comparison

11.1 Earlier claim

The older model stated:

$$Q(x, y) = 4(4x^2 + 11x^2 + 4xy + y^2).$$

This simplifies to:

$$Q(x, y) = 60x^2 + 16xy + 4y^2.$$

The model associated the split:

$$15x^2 = 4x^2 + 11x^2$$

with branchless execution of the six tetrahedral edges.

11.2 Current interpretation

The algebraic identity is exact, but the direct edge mapping was never fully specified.

The newer model separates:


```

six-bit truth table
  selects the active relationship state

four-bit vertex pair mask
  identifies participating vertices

CONS
  preserves ordered construction

quadratic form
  measures, projects, or indexes the construction

```

The form should therefore be treated as a **projection metric or Blackboard index**, not the mechanism that discovers the selected edge.

11.3 Remaining requirement

A complete specification still needs an explicit typed map:

$$\mu : (E_6, V_4, P) \rightarrow Q_e(x, y),$$

where:

- E_6 is the edge-state coordinate;
- V_4 is the vertex-control coordinate;
- P is the plane, Azimuth, or projection profile;
- Q_e is the resulting indexed quadratic projection.

12. Polytope Hierarchy

12.1 Earlier hierarchy

The earlier model used:

```

Tangent tetrahedron
  user graph

Regular tetrahedron
  local bigraph

Stellated tetrahedron
  remote multigraph

```

Triakis tetrahedron
network hypergraph

It also assigned polytope families to tetrahedral vertices:

Cube
Octahedron
Icosahedron
Dodecahedron

and higher-dimensional dual pairs:

8-cell ↔ 16-cell
120-cell ↔ 600-cell

12.2 Current interpretation

These should be treated as **projection lenses**, not literal replacements for tetrahedral vertices.

The tetrahedron remains the self-dual index scaffold.

Two dual families are projected through it:

Cubic/orthoplex family

Cube ↔ Octahedron
8-cell ↔ 16-cell

Icosahedral/dodecahedral family

Icosahedron ↔ Dodecahedron
120-cell ↔ 600-cell

12.3 Antipodal placement

Dual pairs should be placed antipodally across the centroid:

Cube ↔ Octahedron
Icosahedron ↔ Dodecahedron

rather than adjacent merely because they appear consecutively in a list.

12.4 Archimedean and Catalan forms

The intended relation is:

```
Archimedean construction
    primal, often chiral or vertex-oriented view

Catalan dual
    reciprocal face-oriented view

centroid
    common inspection reference

Azimuth
    orientation comparison

difference field
    transformation gauge
```

Snubbed truncation introduces chirality, while the dual construction exposes the corresponding reciprocal structure.

13. Factorial and Clock Interpretation

13.1 Earlier geometric associations

The older model associated:

```
4! = 24
    24-cell/local permutation lens

5! = 120
    5-cell or higher projection lens
```

13.2 Current model

The timing architecture now separates:

```
24
    permutations of FS, GS, RS, US

720
    anchored six-position operational permutations

4,320
    Algorithmic Clock coordinate

5,040
    complete seven-position rotation coordinate
```

The factorial counts should therefore be described as permutation-address resolutions, not as identities equating factorials with polytopes.

A safe statement is:

The 24-cell and 5-cell projection lenses are indexed using permutation resolutions related to $4!$ and $5!$; they are not conventionally identical to those factorial numbers.

14. Geometric Constants and Projection

14.1 Earlier implicit approach

The older geometry-first material sought to derive geometric relationships from polytope incidence rather than relying on stored decimal constants.

14.2 Current proved direction

The current proof architecture makes the boundary explicit:

```
finite incidence
    exact

algebraic or series derivation
    reconstructive

metric projection
    resolution-dependent
```

```
raster output
disposable
```

The tetrahedral incidence supplies the finite route associated with $\sqrt{3}$. The $\{3, 5\}/\{5, 3\}$ incidence route supplies the defining equation for ϕ . The Polybius and diagonal incidence schedules supply the reconstruction route for π .

The π development also supplies an explicit finite approximation bound.

14.3 What changed

The old model described polytope lenses geometrically.

The new model gives them a possible cross-platform role:

Serialize incidence, derivation law, root or phase selector, resolution, and arithmetic profile—
not platform-specific floating-point constants or meshes.

15. Blackboard Pattern

15.1 Earlier understanding

The Blackboard was understood geometrically as:

- a central field;
- four semantic contributors;
- six pairwise channels;
- higher-resolution shape projections;
- a spatial logic index.

15.2 Current interpretation

The Blackboard is now better understood as a tangential block field:

```
vertices or blocks
  retain origin identity

edges
  express relations

centroid
  supplies the shared projective reference
```

```
projection lenses
  expose different spatial resolutions

Delta
  traverses relationship states

CONS
  preserves ordered construction
```

The user's geometry-first interpretation preceded the conventional software-pattern vocabulary, but it corresponds closely to a shared relational workspace.

16. Emergent-Axial-Lisp

16.1 Why it interrupted the geometry work

The geometric model could not become executable until the language layer answered:

- how atoms are represented;
- how CAR and CDR retain order;
- how CONS selects or constructs an edge;
- how a symbolic form becomes a bounded coordinate;
- how evaluation avoids hidden mutable state;
- how `.omi`, `.imo`, and `.o` correspond;
- how axial and tangential positions are selected.

16.2 Current lowering sequence

A normalized lowering pipeline is:

```
source
  (CONS CAR CDR)

semantic binding
  identify block or vertex roles

pair selection
  obtain one weight-two four-bit mask

relation-state update
  update the six-bit edge field

typed transformation
```

```
XOR, shift, rotate, or Azimuth profile
```

```
closure
```

```
  evaluate the relevant face/difference relation
```

```
projection
```

```
  assign clock, rotation, and resolution coordinates
```

```
machine output
```

```
  emit immutable .o representation
```

The Lisp work was therefore not unrelated. It was required to make the tetrahedral spatial model functional.

17. Claims That Should Be Deprecated

The following older claims should not appear unchanged in the final specification:

“Completely closed, frozen, and mathematically locked”

Replace with a status table distinguishing:

```
PROVED  
EXHAUSTIVELY CHECKED  
DEFINED  
IMPLEMENTED  
INTERPRETED  
OPEN
```

“Single-cycle”

Replace with:

Constant bounded lookup with no data-dependent iteration.

Actual cycle count is platform-dependent.

“Physical execution channels”

Replace with:

Finite logical or structural relation channels.

“No sequential loops”

Retain only as a profile restriction where explicitly required. Some proof, tooling, parsing, serialization, or rendering operations may legitimately iterate even when the atomic kernel is branchless.

“O(1) trajectory resolution”

Specify the exact operation whose cost is bounded. Do not apply the claim to an entire scene, proof, or projection pipeline.

“Receipt boundary”

Replace with finite closure or internal relation boundary.

“The quadratic form executes the six edges”

Replace with:

The edge truth table selects relationships; the quadratic form projects or indexes the selected construction.

18. Comparative Status Matrix

Earlier component	Current status	Action
Four semantic vertices	Retained	Freeze exact FS/GS/RS/US binding
Six structural edges	Retained and clarified	Freeze edge order and masks
$2^6 = 64$ truth table	Restored as foundational	Define canonical truth-table order
CONS center	Retained	Define as ordered computational centroid
[6,4,2]	Retained but reclassified	Call architectural incidence profile
[7,4,3]	Retained	Use as compact integrity profile
[8,4,4]	Retained	Use as extended integrity profile
15 formal files	Retained structurally	Model as 5 families × 3 surfaces
24+8 lens	Retained	Treat as geometric terminal view
26+6 lens	Retained	Treat as OMINO operational view

Earlier component	Current status	Action
27+5 lens	Retained	Treat as Tetragrammatron view
28+4 lens	Retained	Treat as four-control/Metatron view
Triakis/Stellated/Regular/Tangent names	Partially retained	Define as named projection profiles
Polytope vertex lists	Interpretive	Convert into typed antipodal dual lenses
Archimedean/Catalan validation	Needs normalization	Define reciprocal incidence comparison
4/11 quadratic split	Algebraically retained	Define exact projection map
STDIN/STDOUT/STDERR planes	Retained with correction	Make STDIN/STDOUT antipodal
<code>0x1D</code> standard-error role	Partially supported	Separate implementation fact from interpretation
Branchless CASE pseudocode	Deprecated	Replace with lookup or mask selection
Single-cycle claim	Deprecated	State bounded complexity only
Receipt language	Deprecated	Use closure coordinates
OMNION name	Deprecated	Normalize to OMINO
Stored geometric constants	Deprecated direction	Derive from incidence and resolution profiles
Emergent-Axial-Lisp	Retained as necessary	Complete formal lowering rules

19. Recommended Canonical Layering

The final specification should be organized into six layers.

Layer 1 — Tetrahedral semantic incidence

```

RULES
FACTS
CLOSURES
COMBINATORS
CONS centroid

```

Layer 2 — Tetrahedral logic field

four-bit vertex masks
six edge indices
six-bit relationship states
64-row truth-table coordinate

Layer 3 — Nested gauge integrity

OMINO [6,4,2]
OMICRON [7,4,3]
OMNICRON [8,4,4]

Layer 4 — Addressing and projection lenses

32-position lenses
low/high control planes
axial origins
tangential completions
Metatron
Tetragrammatron
Polybius
Azimuth

Layer 5 — Temporal and rotational execution

Delta
Algorithmic Clock
rotation coordinate
deterministic replay

Layer 6 — Polyharmonic visual construction

incidence-derived geometry
dynamic resolution
polytope dual lenses
2D and 3D block composition
cross-platform reconstruction profile

Conclusion

The old and new versions are not fundamentally competing designs.

The older version contains the **geometric generative model**:

4 vertices $\rightarrow$ 6 relations $\rightarrow$ 64 truth states $\rightarrow$ CONS-centered spatial construction.

The newer version contains the **normalization and execution discipline**:

OMINO₆₄₂ $\rightarrow$ OMICRON₇₄₃ $\rightarrow$ OMNICRON₈₄₄ $\rightarrow$ bounded addressing $\rightarrow$ deterministic projection and replay.

The correct integration is therefore:

The tetrahedral truth-table field is the logical foundation. OMINO [6,4,2] is its reduced operational gauge. OMICRON and OMNICRON add compact and extended integrity. CONS preserves ordered construction at the centroid. The quadratic form, polytope duals, Polybius index, Azimuth, clock, and rotation coordinates project the same finite incidence into progressively higher-resolution Blackboard views. Emergent-Axial-Lisp provides the missing executable lowering from symbolic relations into those finite spatial coordinates.

The remaining work is not to invent another architecture. It is to freeze the exact mappings between layers while preserving which claims are proved, implemented, defined, interpreted, or still open.